

RESEARCH ARTICLE

Multimodal Hydrogel Bioelectronics for Closed-Loop Diagnosis and Treatment of Cardiac Diseases

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ABSTRACT

The development of precise in situ cardiac monitoring and effective therapeutic management systems is crucial for advancing cardiovascular pathophysiology understanding and improving diagnostic-therapeutic integration. Current bioelectronic devices face fundamental limitations in establishing compliant tissue interfaces, hindering accurate spatiotemporal mapping of myocardial strain/electrophysiology and manipulation of cardiac electromechanical activity. To address this, we engineered a multimodal hydrogel multielectrode array (MHMEA) using integrated multimaterial 3D printing, featuring: tissue-matching mechanical compliance (Young's modulus of 430 kPa) enabled by the hydrogel construction; instant tough bioadhesion (interfacial toughness of 100 J m⁻²) through dynamic covalent crosslinking; and triple-functional conductive hydrogel electrodes for concurrent strain sensing, electrophysiological mapping, and electrical stimulation. This synergistic design achieves stable interfacial integration with dynamically beating myocardium, enabling in situ synchronous high-resolution mapping of myocardial strain and epicardial electrophysiology for precise pathological diagnosis (e.g., myocardial infarction). Furthermore, the platform delivers targeted electrical stimulation therapy, establishing a closed-loop diagnostic-therapeutic system for personalized cardiac disease management through electromechanical synchronization.

1 | Introduction

Cardiovascular diseases (CVDs) constitute a pressing global health hazard, accounting for nearly one-third of all deaths globally [1, 2]. Among these, myocardial infarction (MI), caused by occlusion of the coronary arteries, is particularly severe due to its sudden onset, complex sequelae, and high fatality rate [1, 2]. A significant impediment to addressing these challenges lies in the scarcity of advanced tools capable of spatiotemporal mapping and manipulating cardiac metabolic and electromechanical activity [3]. This limitation hampers our ability to delineate the complex

pathophysiological processes underlying heart disease, monitor intraoperative and postoperative recovery, and devise effective and timely clinical interventions [3–5]. Therefore, the innovation of sophisticated technologies capable of spatiotemporal mapping and manipulation of cardiac metabolic and electromechanical functions is pivotal for refining diagnostic accuracy, informing treatment protocols, and ultimately elevating the standard of patient care and outcomes [3, 6–8].

Advancements in micro-/nanofabrication technologies have driven the creation of sophisticated bioelectronic devices, offering

precise interfacing with the human body [7, 9, 10]. Multielectrode arrays (MEAs) are particularly promising for the high-fidelity recording and mapping of extracellular electrical potentials, crucial for understanding cardiac excitation wave propagation and identifying arrhythmia substrates [6, 7, 11]. However, the rigidity of these devices contrasts with the soft, dynamic nature of heart tissue, potentially leading to damage, clot formation, arrhythmia, and device failure [1, 12]. While special structural designs have enhanced the flexibility and stretchability of MEAs, they still fall short in synchronizing with the heart's movements, and rigid components can exert harmful stress on cardiac tissue [11, 13, 14]. The emergence of intrinsically stretchable array electrodes offers a solution to improving the interfacial interfacing between bioelectronic devices and cardiomyocytes [15, 16]. However, the incorporation of significant rigid conducting fillers in elastomers with low toughness gives rise to a disparity in mechanical stiffness and diminished tissue compatibility, and these dry electrodes may also exhibit suboptimal electrochemical performance in the hydrated physiological environment [17–19]. An alternative approach involves leveraging soft, wet, ionically conducting, and biocompatible functional hydrogel materials to fabricate those array electrodes [20–23]. Conducting polymer hydrogels (CPHs) emerge as superior material platforms, harmoniously merging the softness and high-water content attributes of hydrogels with the unique ionic-electronic conductivity of conducting polymers [20, 21, 24]. Bioelectronic devices made from CPHs enable soft mechanical coupling and efficient bidirectional electrical interfacing with living organisms, concurrently maintaining minimal tissue disruption and negligible immune responses [22, 25]. Despite the progress in high-precision manufacturing of CPHs-based electrode arrays utilizing sophisticated techniques such as photolithography, laser-induced forward transfer, and 3D printing, these devices are currently inadequate in delivering comprehensive multimodal physiological monitoring and therapeutic interventions [22, 26–29]. This limitation constrains the versatility of CPHs-based bioelectronics in addressing the complexities of various diseases and disorders. To broaden the utility of CPHs-based bioelectronics in healthcare monitoring and biomedical theragnostics, there is an urgent requirement to innovate hydrogel bioelectronics with multifunctional capabilities.

Herein, we present a multimodal hydrogel multielectrode array (MHMEA) engineered via integrated multimaterial 3D printing to enable closed-loop diagnosis and therapy for cardiac diseases. Its tripartite architecture comprises (i) a hydrogel adhesive layer ensuring seamless epicardial integration, (ii) conducting polymer hydrogel electrodes performing concurrent strain sensing, electrophysiological mapping, and electrical stimulation, and (iii) an insulating hydrogel encapsulation providing mechanical protection. This design achieves tissue-level compliance and stable interfacing with dynamic myocardial tissue, permitting in situ, simultaneous high-precision spatiotemporal mapping of myocardial strain and epicardial electrophysiology. By precisely locating pathological regions and delivering targeted electrical stimulation, the MHMEA establishes a self-regulating therapeutic cycle that couples real-time diagnosis with immediate intervention.

2 | Results

2.1 | Design of Multimodal Hydrogel Bioelectronics

Bioelectronic devices with multifunctionality hold immense significance for comprehensive healthcare monitoring and treatment, as they integrate a spectrum of recording, self-sensing, and stimulating capabilities [30, 31]. Here, MHMEA is fabricated through multi-material 3D printing technology, which shows the ability to simultaneously monitor electrocardiogram (ECG) signals and myocardial strain, as well as provide electrical stimulation functionality (Figure 1a). This MHMEA is constructed entirely from hydrogel precursor inks, encompassing adhesive hydrogels for seamless interfacing with the epicardium, CPHs for strain sensing, electrophysiological recording, and electrical stimulation, as well as insulating hydrogels for encapsulation. We have characterized the chemical structure of the PAA-NHS copolymer by ^1H NMR (Figure S1), corroborating the presence of functional NHS and acrylate moieties. These meticulously formulated hydrogel precursor inks, exhibiting excellent shear-thinning properties, facilitate their precise extrusion printing into predetermined 2D and 3D patterns and structures (Figure S2). In brief, the fabrication process commences with the printing of a substrate layer composed of a non-conducting bioadhesive hydrogel, followed by printing the functional layers (i.e., strain sensing, recording, and stimulating electrodes) using CPHs inks, and is finalized with an encapsulation layer using an insulating hydrogel. Notably, an insulating hydrogel layer is also incorporated between the strain-sensing and the electrophysiological recording/stimulation layers during the fabrication process to avert signal crosstalk. Detailed information regarding the fabrication of the MHMEA is provided in the Experimental Part and Supporting Information. In light of their submillimeter ($\sim 200\ \mu\text{m}$) thickness, tissue-like mechanics (Young's modulus of 430 kPa) (Figure 2d; Figure S3), and exceptional adhesion properties, the as-prepared MHMEA is capable of establishing a conformal and robust interface with wet biological tissues and dynamically moving organs. Owing to the seamless integration of MHMEA with tissues/organs, the device can accurately capture physiological signals and efficiently execute electrical stimulation for modulating life activities (Figure 1b,c).

2.2 | Electrical, Bioadhesion, and Mechanical Performance of the MHMEA

The MHMEA attained swelling equilibrium within 2 h in PBS (37°C), with out-of-plane (thickness) swelling reaching 140%, contrasting with the sub-115% in-plane swelling (length and width) (Figure S4). This controlled anisotropy is critical for ensuring stable device-tissue integration and consistent mechanical compliance during implantation. Figure 2a exhibits various mechanical deformations of the MHMEA, including bending, crumpling, and stretching. The MHMEA preserved its structural integrity without delaminating when subjected to these deformations, attributable to the matched mechanical properties (Figures S5–S7) and the robust chemical crosslinking across the

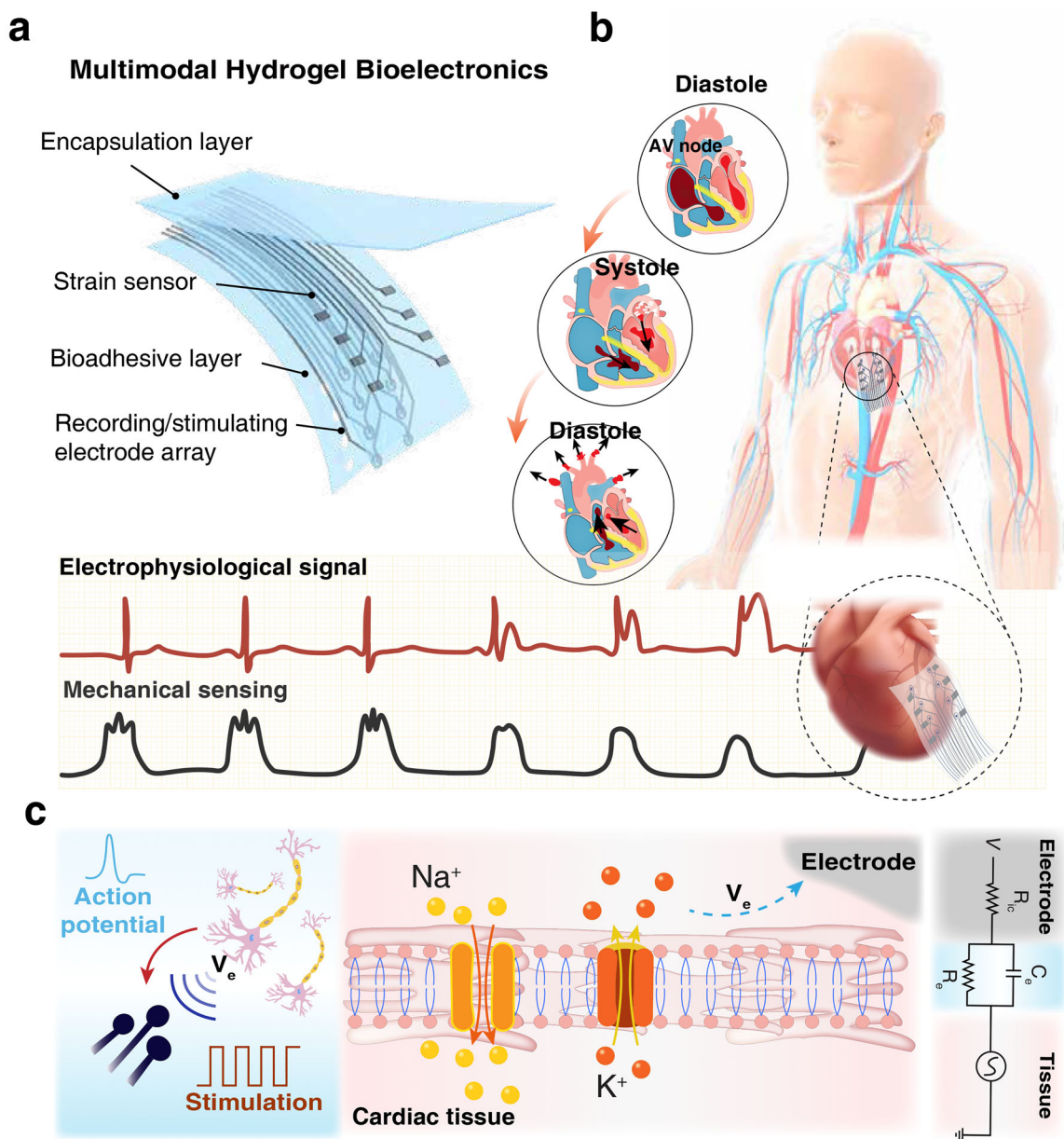


FIGURE 1 | Design of multimodal hydrogel multielectrode array (MHMEA). (a) Schematic illustration of the MHMEA composed of adhesive hydrogels for robust interfacing with epicardium, CPHs for strain sensing, electrophysiological recording, and electrical stimulation, along with insulating hydrogels for protective encapsulation. (b) Schematic diagram of the MHMEA for in situ closed-loop diagnosis and treatment of cardiac diseases by in situ self-sensing, epicardial electrophysiological mapping, and electrical stimulation. (c) Schematic diagram showing the mechanism of biopotential signaling detected by the MHMEA, which is generated by excitable cells through ion transport across membranes.

various layers. The interfacial chain entanglement, coupled with the formation of stable covalent bonds, confers an interfacial toughness exceeding 100 J m^{-2} and a shear strength of over 100 kPa between the functional layer and substrate/encapsulation layer of the MHMEA (Figure S8). Moreover, the stable long-term interfacial adhesion ($>120 \text{ kPa}$) enables the MHMEA device to resist the dynamic moist in vivo environment and maintain its structural integrity throughout prolonged implantation. (Figure S9) The Scanning Electron Microscopy (SEM) image also clearly reveals a coherent adhesion at each heterogeneous interface within the multi-layered hydrogel bioelectronics (Figure S10). To evaluate the adhesion performance of the MHMEA with biological tissue (i.e., porcine skin), we performed 180° peeling

tests and lap-shear tests to quantify the interfacial toughness and shear strength, respectively. The interfacial toughness between hydrogel bioelectronics and porcine skin is higher than 100 J m^{-2} and the shear strength of more than 100 kPa , which is obviously better than the control sample s with PAA (w/o NHS moieties) instead of PAA-NHS (interfacial toughness $\sim 20 \text{ J m}^{-2}$; shear strength $\sim 40 \text{ kPa}$) (Figure S11). As shown in Figure 2b, the MHMEA could robustly adhere onto the chicken heart, providing a seamless and robust integration even under mechanical deformation. We also probed the stability of adhered joints under dynamic conditions by exposing them to a cyclic abrasion measurement (more details in the Experimental Section) for 1000 cycles, and a slight change in interfacial toughness and shear

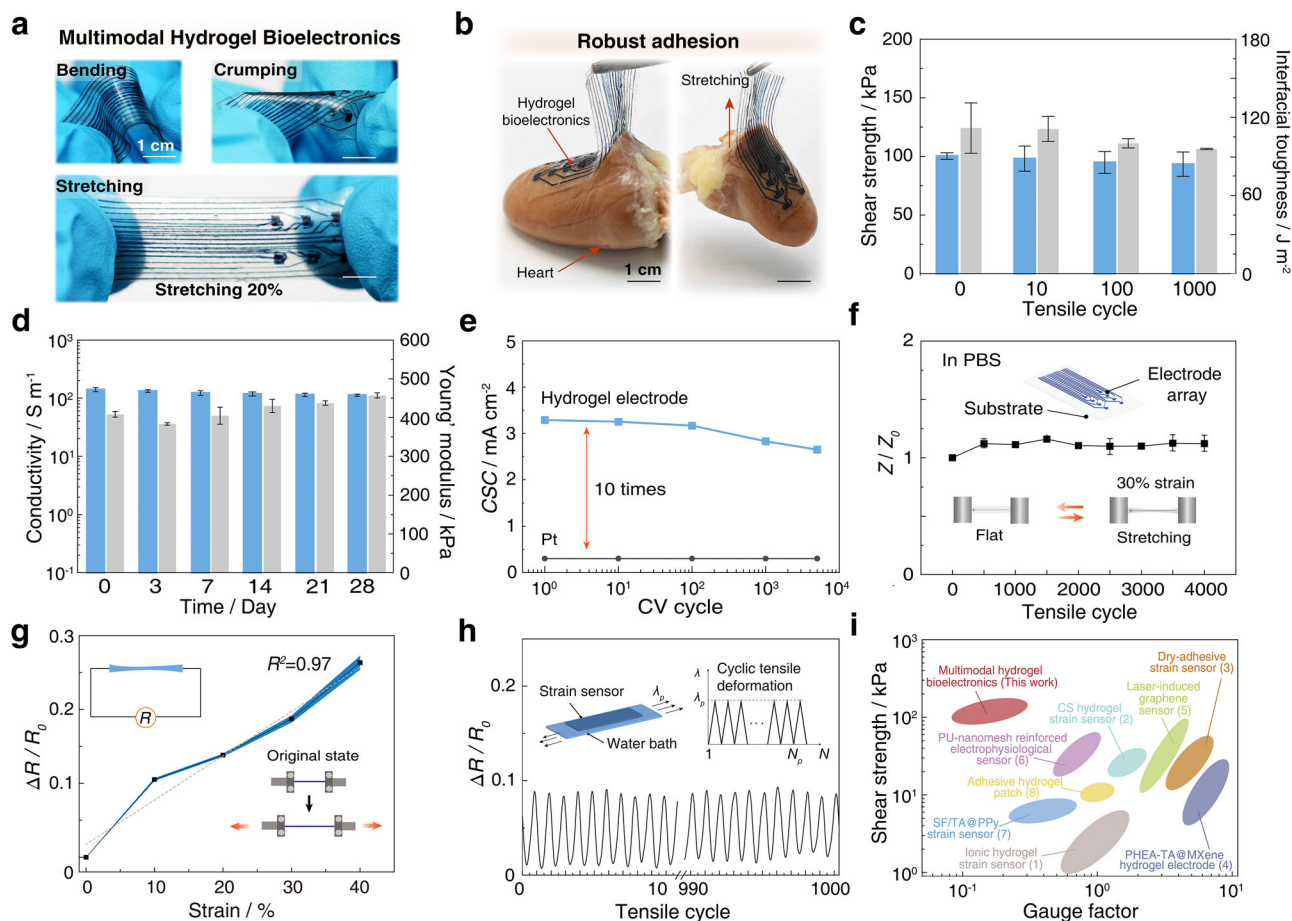


FIGURE 2 | Characterization of the MHMEA. (a) Various mechanical deformations of the MHMEA, including bending, crumpling, and stretching. Scale bar: 1 cm. (b) Images corroborating the mechanically conformal adhesion between the MHMEA and heart, as well as robust adhesion without failure upon stretching. Scale bar: 1 cm. (c) Evolution of adhesion performance (interfacial toughness and shear strength) during the 1000-cycle cyclic stretching at a strain of 30%. (d) Conductivity and Young's modulus of the CPHs samples in PBS over 28 days at 37°C. (e) The CPHs-based electrode can provide a favorable charge storage capacity (CSC) versus cyclic voltammetry (CV) over 10 times higher than the Pt electrode. (f) Impedance of CPHs-based electrode as a function of stretching cycles at a strain of 30%. Z and Z_0 represent the impedance value at a specific stretching cycle and the initial impedance, respectively. (g) The gauge factor of the strain sensor at different strains. (h) Dynamic relative resistance change ($\Delta R/R_0$) of the CPHs-based strain sensors during 1000 cycles of stretching and relaxing. (i) Comparison chart by plotting the gauge factor and adhesion strength of recently reported CPHs used in bioelectronics. Data in c, d, e, f, g are means $\pm$ SD, $n = 3$.

strength is observed, corroborating the robust anchorage of our MHMEA on tissues/organs (Figure 2c).

Subsequently, the electrical, electrochemical, and strain-sensing attributes of CPHs as integral functional elements within MHMEA were systematically evaluated. The CPHs exhibit a superior electrical conductivity of $>100 \text{ S m}^{-1}$, which is substantially higher than that of insulating hydrogels used in encapsulation layers ($< 0.01 \text{ S m}^{-1}$) (Figure S12) and biological tissues ($0.3\text{--}0.7 \text{ S m}^{-1}$). These CPHs maintain stable electrical conductivity in a physiological environment (incubation in PBS at 37°C) for more than 28 days, thereby satisfying the requirements for prolonged in vivo applications (Figure 2d). Furthermore, after 1000 stretching cycles at 40% strain, the CPHs demonstrate exceptional stability with a negligible decrease in conductivity (Figure S13), confirming its outstanding reliability for dynamic bioelectronic interfaces. Moreover, the CPHs-based electrode demonstrates a superior charge storage capacity (CSC) of $\sim 3 \text{ mC cm}^{-2}$ and a low impedance of $\sim 200 \Omega$ (at 1 kHz),

and a nearly uniform CSC from 8-independent channels within MHMEA is maintained, guaranteeing consistent and reliable performance throughout this system (Figure 2e; Figure S14). The charge injection capability (CIC) of the CPHs-based electrode was characterized to evaluate its effect on the electrical stimulation efficacy of MHMEA. The CPHs exhibit a significantly higher CIC ($\sim 2.0 \text{ mC cm}^{-2}$) compared to conventional metal electrodes ($< 0.05 \text{ mC cm}^{-2}$) [26], and this superior CIC was maintained with stability after exceeding 10^5 cycles (Figures S15). Figure 2f shows that the MHMEA maintains a stable electrochemical performance after 4000 stretching against 30% strain, suggesting the excellent mechanical stability of the devices.

We further conducted a thorough assessment of the strain-sensing capabilities of the CPHs integrated within the MHMEA. To measure their strain-sensing performance, we applied precise loads spanning a strain interval from 10% to 40%, utilizing an experimental apparatus that included a biometric tension device (Cellscale) coupled with an LCR meter for resistance testing

(Figure 2g; Figure S16). The results show that the response to the applied strain was distinguished clearly as a step-like response, indicating high sensitivity and reliability of the sensor in detecting mechanical deformation. Additionally, the CPHs demonstrate a swift response time of approximately 100 ms, and upon load removal, the resistance rapidly returns to its initial value within 120 ms, with negligible hysteresis throughout the strain cycle (Figure S17). Moreover, the 8-independent channels within the MHMEA exhibited a high degree of consistency (Figure S18). Despite being subjected to very subtle strains (1%), the CPHs still manifest an apparent change in electrical resistance, a characteristic that is highly beneficial for the precise monitoring of cardiac deformations, thereby enhancing the diagnostic capabilities of MHMEA (Figure S19) [32]. A long-term strain-sensing measurement over 1000 cycles at a 40% strain confirms the enduring stability and reliable sensing performance of these CPHs-based strain sensors within the MHMEA (Figure 2h; Figure S20). We compared two key parameters (i.e., adhesion performance and gauge factor) with other conducting hydrogels used in bioelectronics. Our designed adherable MHMEA exhibits superior bioadhesion performance compared to most existing conducting hydrogel-based bioelectronics (Figure 2i; Table S1). This exceptional performance ensures reliable and stable signal transmission between MHMEA and cardiac tissue [33–40].

2.3 | Mechanical Compliance of the MHMEA Biointerface

A key feature of this MHMEA is its capacity to maintain a stable mechanical interface with the tissue while exerting minimal force on the myocardium during both contraction and relaxation. Through 3D finite element methods (FEM) and quantitative analysis, the stress exerted by our hydrogel bioelectronics on the epicardium is evaluated, providing insights into the stability of the electrode–tissue interface (Figure S21). Figure 3a demonstrates that the MHMEA remains closely attached to the surface of the heart during both systole and diastole. This conformability is crucial for the electrode's functionality, as it must maintain intimate contact with the heart while accommodating the dynamic changes in its shape and size during each cardiac cycle. During cardiac contraction and relaxation, stress is generated at the interface between the electrode and the heart. To prevent the MHMEA from damaging cardiac tissue, it is essential to ensure the stress remains within safe limits. Finite element analysis (FEA) was used to calculate the radial pressure exerted by the electrode under varying shear modulus conditions. Results indicate that the radial pressure applied by the electrode consistently remains lower than the intrinsic pressure of the epicardium (Figure 3b,c). This demonstrates the electrode's superior mechanical interface stability with the heart, without imposing excessive stress that could impair cardiac function. The thickness of the electrode alters the mechanical interface between the electrode and the heart. Theoretically, a thicker electrode could exert greater stress or resistance on cardiac motion, potentially interfering with normal heartbeats. However, analysis reveals that even with variations in thickness, the radial pressure exerted by the electrode remains below the epicardial pressure (Figure 3d,e). This indicates that the electrode does not impede the heart's natural contraction and relaxation, maintaining its functional compatibility. Therefore, the robust anchored interface

formation of the MHMEA to the heart tissue ensures that the devices do not interfere with the normal physiological function of the heart, which is particularly important for biomedical implants and monitoring systems.

2.4 | In Vivo Cardiac Physiological Measurement and Modulation

Given that some severe cardiac diseases (e.g., myocardial infarction) can occur on the cardiac surface, in situ cardiac physiological monitoring is of paramount importance for the accurate diagnosis and effective treatment of cardiac conditions [19, 41]. Multimodal bioelectronics offer the potential for simultaneous monitoring and therapeutic intervention, thereby enhancing the management of cardiac pathologies [30, 31]. Our MHMEA was successfully implanted onto the surface of a healthy rat heart in vivo, as depicted in Figure 4a. Employing a dry-crosslinking mechanism, the MHMEA rapidly achieved bioadhesion with the cardiac tissues (Figure 4b), facilitating seamless integration and stable contact for physiological measurement and modulation. As illustrated in Figure 4c, each of the eight independent channels within the MHMEA is capable of accurately capturing normal cardiac electrical signals, in which the distinct waveforms corresponding to atrial excitation (P wave), ventricular excitation (QRS complex), and ventricular relaxation (T wave) were orderly and alternately distinguished. Spatiotemporal epicardial-potential maps are established based on the reliable operations of each of the sensing channels within the MHMEA, which reveal the horizontal propagation of cardiac activation from the left to the right during a sinus rhythm (Figure 4d). We further used MHMEA to monitor the changes in myocardial strain during cardiac beating. Thanks to the strong interfacial anchorage, all 8-independent strain sensors within the MHMEA can capture myocardial strain in different areas of the heart (Figure 4e). The resistance peaks recorded by the strain sensors were in close temporal alignment with the waves of the electrocardiograms (ECGs), where each characteristic peak corresponds to the corresponding wave patterns within the ECG traces [32]. Additionally, the spatiotemporal mapping of the strain distribution also validated that the strain trace was in alignment with the ECG trace, confirming the correspondence of the heart rate as indicated by both modalities (Figure 4f).

For the electrical pacing case, a stimulus node is strategically positioned close to each electrophysiological recording site to enable precise electrical modulation of the targeted diseased area. As shown in Figure 4g, electrical stimulation has emerged as a validated technique for managing heart disorders related to abnormal electrical conduction, including atrioventricular block, arrhythmias, and myocardial infarction, etc. [21, 42, 43]. We conducted electrophysiological stimulation of the heart utilizing the stimulating electrodes integrated within the MHMEA and concurrently recorded changes in the ECG signals and cardiac deformation using the dedicated recording and strain-sensing electrodes. As shown in Figure 4h, the electric stimulation (0.5 V at a frequency of 8 Hz and a duration of 4 ms) led to synchronous enhancement in strain period and heart rate, confirming that our MHMEA can manipulate the rate and rhythm of a heartbeat as a pacemaker. As shown in Figure 4i,k, measurements of epicardial ECG revealed apparent pacing spikes with a frequent interval

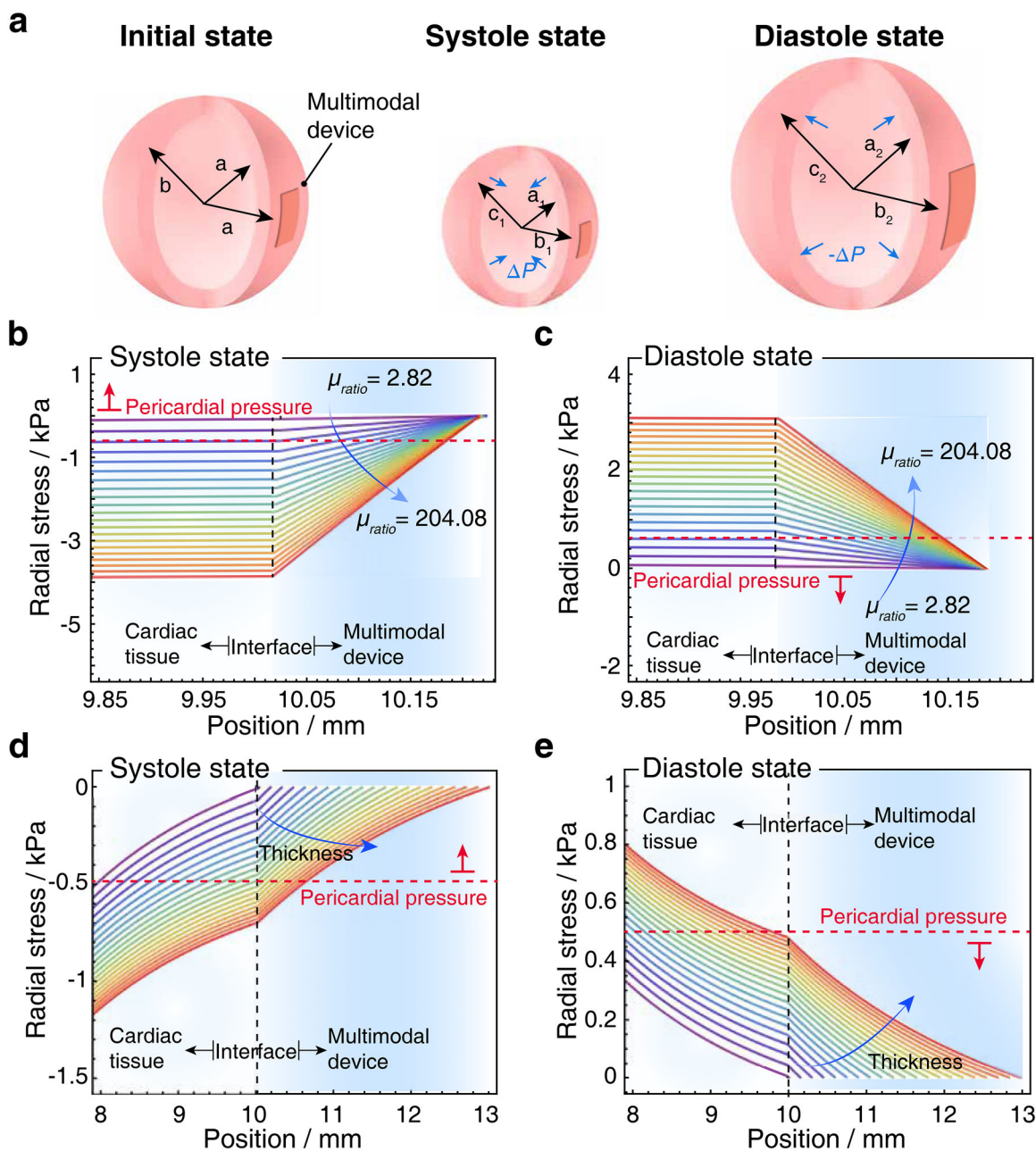


FIGURE 3 | Analysis of radial stress at the interface between the epicardium and the integration of the MHMEA. (a) Schematic diagram of the formation of a stable interface of the MHMEA on the surface of the heart during the cardiac systole and diastole state. (b, c) Under different shear modulus conditions of the electrode, during the contraction and relaxation of the heart, the radial stress at the interface between the MHMEA and the heart changes. Adjusting the mechanical modulus of MHMEA, the radial pressure at the interface between our MHMEA and the heart changes during the cardiac systole state (b) and diastole state (c), but the interface remains stable, indicating that the device is mechanically compatible with the heart. (d, e) Under different device thickness changes, during the heart's contraction and relaxation, the radial stress at the interface between the MHMEA and the heart varies in response to changes in device thickness. Specifically, the radial pressure at this interface fluctuates during diastole (d) and systole (e), adapting to the dynamic conditions of the cardiac cycle. Importantly, the radial stress at the interface remains below epicardial pressure, ensuring that the MHMEA does not interfere with the heart's normal physiological functions. This allows the device to adhere securely to the heart surface without imposing any restrictive effects on cardiac movement.

(~420 bpm) and shrinking peak-to-peak interval (0.2–0.45 ms) and slightly amplified QRS complex in contrast to the sinusoidal rhythm without electrical stimulation (300–320 bpm), and this process does not affect the monitoring signal-to-noise ratio (SNR). Meanwhile, the strain sensors exhibited a consistent change trend

(Figure 4j), confirming the absence of signal crosstalk among the different electrodes.

Additionally, the integration of MHMEA with the epicardium could trigger a physiological response from the body,

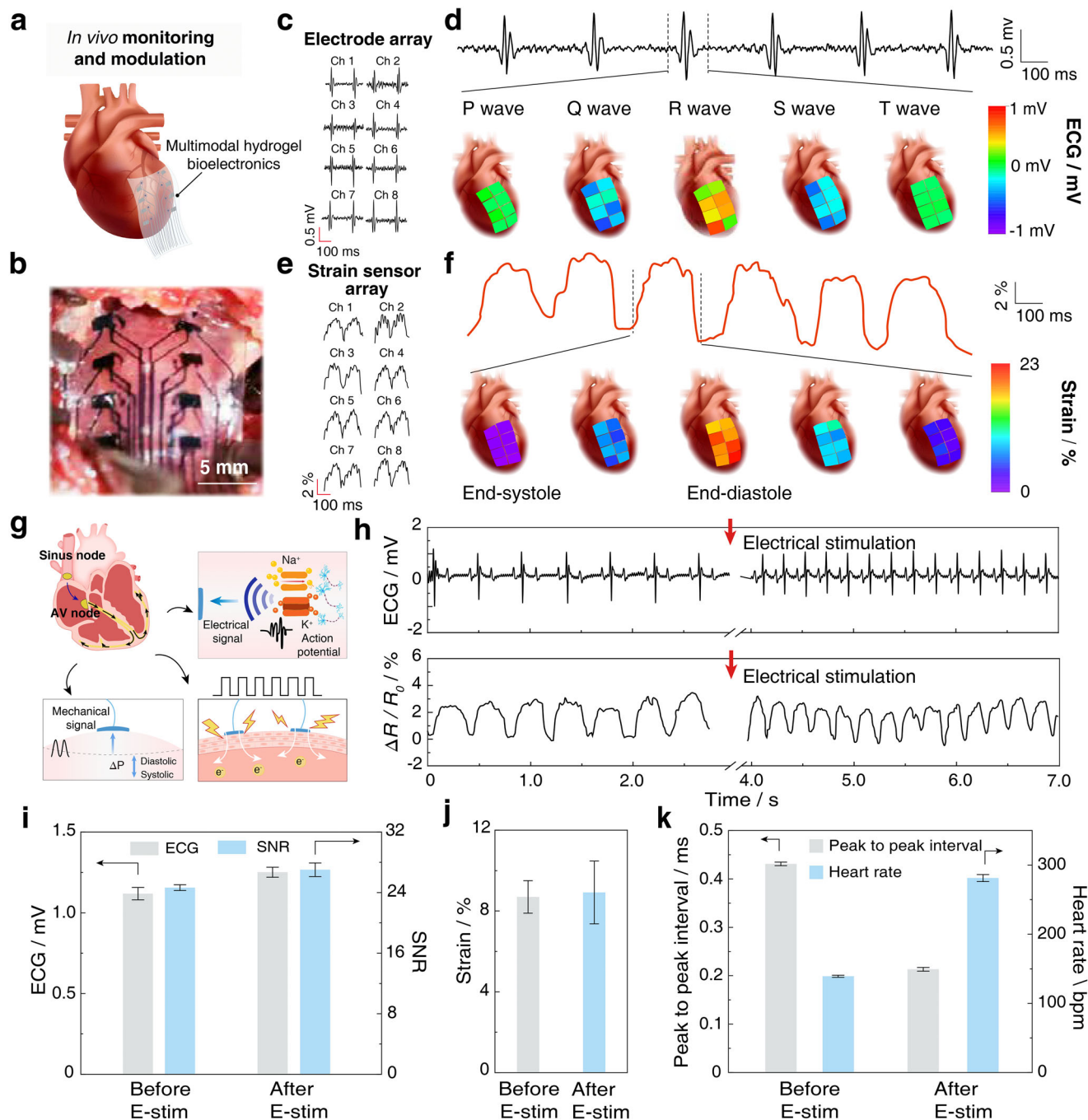


FIGURE 4 | In situ monitoring and modulation of cardiac beating motion in a live rat model. (a) Schematic illustration of MHMEA for in situ monitoring of the physiological signals in a living rat model. (b) Image of an MHMEA on the epicardial surface of a living rat heart. Scale bar: 1 cm. (c) ECG traces from MHMEA. (d) Spatiotemporal biopotential mapping results from 8 individual channels at five waves (P, Q, R, S, and T waves) of the healthy rat heart rhythm. (e) Strain curve traces from 8 individual strain sensors within MHMEA. (f) Spatiotemporal mapping of epicardial strain distribution was measured using this MHMEA during a single cardiac cycle of a normal anesthetized rat. (g) The schematic illustrates the use of MHMEA to record cardiac activity in a live rat model. The mechanical beating of the heart causes changes in electrode resistance, while bioelectric signals from electrically active cells propagate toward the electrode, allowing for electrophysiological monitoring. During stimulation, the electrode applies an electric potential, injecting ionic currents into the tissue medium and generating a corresponding electrical potential in the target tissue. (h) Representative ECG and strain curve trace from the MHMEA with and without electrical stimulation. (i–k) Statistical analysis of ECG amplitude and SNR (i), epicardial strain (j), heart rates, and the peak-to-peak interval (k) of the rat's cardiac activity before and after electrical stimulation. Data in i, j, k are means $\pm$ SD, $n = 3$.

necessitating a rigorous evaluation of the implant's biocompatibility. A comprehensive suite of *in vitro* and *in vivo* assays was conducted using a rat model, encompassing assessments of cell viability (Figure S22), inflammatory response (Figure S23), and whole blood analysis (Figures S24 and S25). Following a 5-day incubation period, the cell culture medium containing the hydrogel electrodes exhibited *in vitro* cytotoxicity levels equivalent to those of the control in rat cardiomyocytes. Subsequent to subcutaneous implantation in rats, a 4-week evaluation of the inflammatory response was undertaken using hematoxylin and eosin (H&E) staining and macrophage enumeration. Figure S26 reveals the absence of significant inflammation in the subcutaneous tissue surrounding the implant, indicating a favorable biocompatibility profile. Moreover, the whole blood analysis demonstrated no adverse effects on blood parameters, suggesting that the hydrogel electrodes do not elicit a systemic immunological response. These findings collectively support the suitability of the MHMEA for epicardial applications, with negligible immune reaction and maintained myocardial functionality.

2.5 | In Situ Closed-Loop Diagnosis and Treatment

Despite ECG monitoring remaining the most frequently utilized method for assessing cardiac activity in clinical settings, reliance on a single ECG monitoring may result in misdiagnosis [12, 32]. For instance, ventricular hypertrophy may exhibit ECG patterns that closely mimic those of an MI, both featuring an apparent ST-segment elevation (Figure 5a) [44–46]. Consequently, a multifaceted diagnostic approach is essential, necessitating the integration of additional modalities such as histological assessment, echocardiography, and clinical angiography to comprehensively evaluate potential cardiac conditions [47, 48]. Nevertheless, these techniques usually fall short of achieving precise localization of the lesion site and do not offer site-specific therapeutic interventions [32]. Our MHMEA provides real-time monitoring with high spatial resolution and targeted therapy therapeutic capacity, thereby promising to significantly enhance diagnostic precision and therapeutic efficacy in cardiac care. In order to verify the practical application efficacy of MHMEA, two disease models of ventricular hypertrophy and MI were constructed, respectively (Experimental Part and Figure S27). The echocardiogram results confirmed the presence of both systolic and diastolic dysfunction subsequent to myocardial infarction (Figure 5b, top), characterized by reduced ejection fraction and impaired filling of the ventricles (Figure S28c,d) [32]. Additionally, ventricular hypertrophy was evident through the observation of a significant thickening of the heart wall (Figure 5b, bottom), and accompanied by an increase in ejection fraction and fraction shorten (Figure S28) [48].

The ECG results obtained through the MHMEA showed an apparent elevation of the S-T segment in the hearts of individuals with MI and ventricular hypertrophy (Figure 5c; Figures S29 and S30). Moreover, the significant increase in QRS interval duration also indicates the presence of intraventricular block or ventricular hypertrophy (Figure 5d). Based on relative activation delay times, the longitudinal (top-to-bottom) and transverse (left-to-right) conduction velocities in the left ventricle of MI rats were 0.27 and 0.16 m s⁻¹ (Figure 5e,f), respectively, while in ventricular

hypertrophy rats, the conduction velocity was 0.42 and 0.40 m s⁻¹ (Figure 5g,h). The results clearly revealed that the signal was significantly different from that of healthy rats (Figures S31 and S32), and the signal propagation was blocked, which was consistent with previous cardiac electrophysiology studies [6]. It is evident that both diseases elicit comparable electrophysiological signal transduction characteristics, rendering a sole ECG examination insufficient for providing a definitive diagnostic conclusion. Owing to the seamless integration of MHMEA with cardiac tissue, the MHMEA is capable of precisely detecting the strain of the myocardial tissue throughout the heartbeats. Analyses of the spatiotemporal mapping of the strain distribution within heart tissue revealed pronounced differences in stress patterns between hearts affected by MI and left ventricular hypertrophy (Figure 5j–m). As shown in Figure 5k–n, the strain amplitude in the infarction area was notably reduced to approximately 2%, significantly lower compared to the strain observed in both the normal group (~15%) and the left ventricular hypertrophy group (~8%). This diminished strain is attributed to the presence of fibrotic tissue that develops in the infarcted region, which impairs the tissue's ability to deform and contribute to cardiac contraction.

We further demonstrated the utility of our MHMEA for *in situ* diagnostics and therapeutic intervention in a rat model of MI. Initially, local MI was induced in the rats by occlusion of the left anterior descending (LAD) coronary artery, and a whitish discoloration was observed at the site of cardiac ligation (Figure S33). As depicted in Figure 6a, electrophysiological recordings within the MHMEA detected MI, characterized by significantly abnormal ECG patterns. Previous reports have extensively demonstrated that electrical stimulation is an effective strategy for myocardial repair [31, 40]. It is well established that electrical stimulation augments cardiomyocyte function by upregulating genes critical for cardiac contraction, calcium handling, and anti-apoptotic pathways, while simultaneously reinforcing intercellular electrical coupling via increased Connexin 43 (CX43) expression to mitigate apoptosis [43, 49]. Furthermore, electrical stimulation promotes angiogenesis via VEGF upregulation, facilitating endothelial cell proliferation and neovascularization. Crucially, the therapeutic efficacy of these reparative processes is significantly influenced by the specific stimulation parameters employed. In response, the targeted electrical stimulation (50 mV, 8 Hz frequency, 4 ms pulse duration) was administered to modulate the abnormal cardiac activity, in accordance with the previous protocols [50–52]. Seven days post-MI, MHMEA was implanted into the infarcted regions of the MI group, the sham-operated group (representing healthy hearts), and the MI control group (without electrical stimulation) for comparison.

We further investigated the capability of MHMEA for long-term *in vivo* cardiac modulation. Electrical stimulation was administered at seven-day intervals, and subsequent cardiac function was monitored by echocardiography at the 4-week endpoint (Figure 6b). Although the formulation of MHMEA incorporates minor fractions of biodegradable polymers, such as CTS and HEC, its structural integrity is predominantly governed by a chemically cross-linked, non-degradable network of PEDOT:PSS and PAA. This network architecture dictates an exceptionally slow overall degradation kinetic with a bulk degradation timeline typically spanning several months [53, 54];

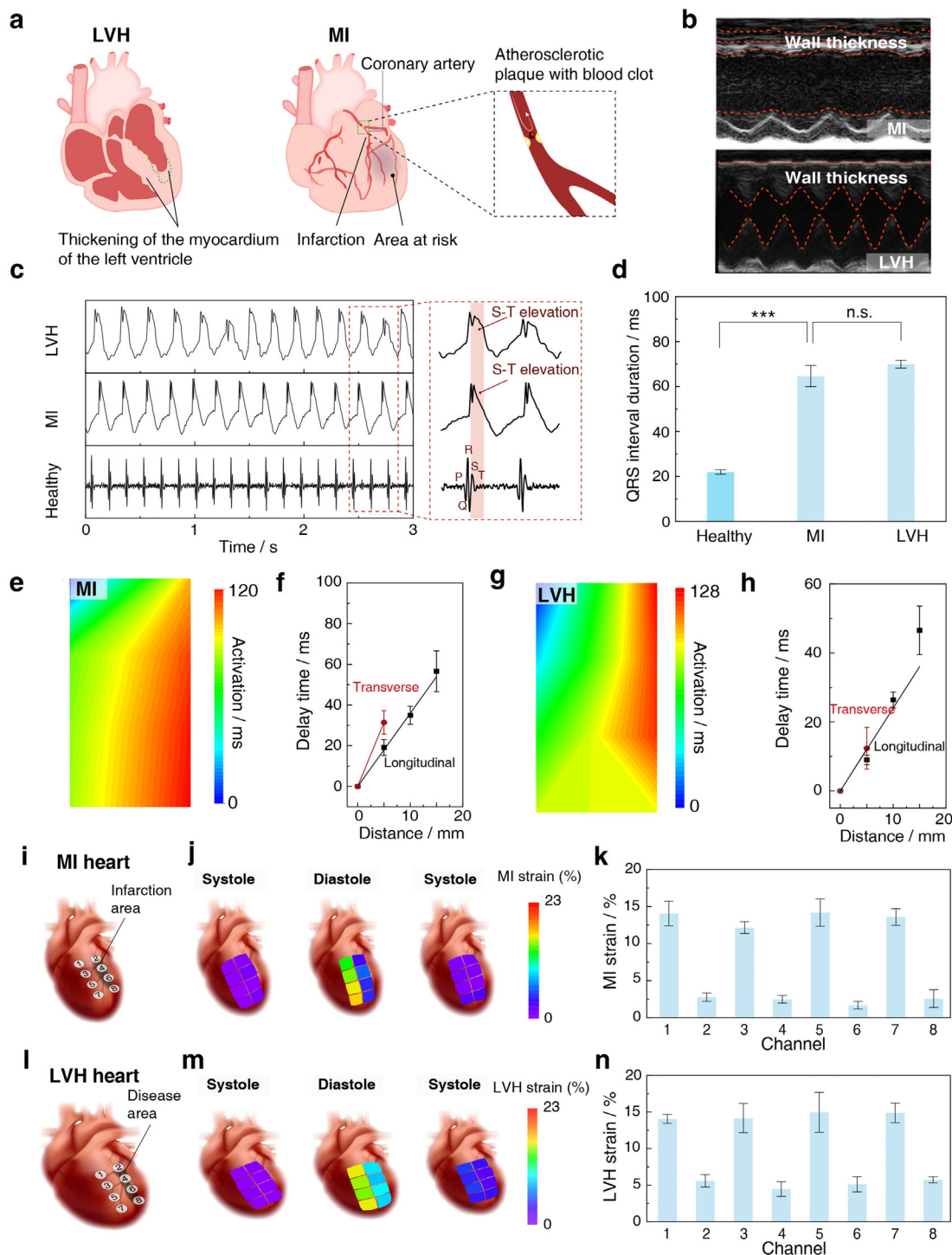


FIGURE 5 | In vivo validation of MHMEA for cardiac disease monitoring. (a) Schematic illustration of the normal and diseased (MI and LVH) cardiac beat movement in a living rat model. (b) Echocardiographic evaluation of left ventricular function across different groups. (c) Representative measurements of local cardiac contractions using the MHMEA in a living rat heart. Both LVH and MI rats have elevated ST segments in ECG signals. (d) Statistical analysis of QRS interval duration for different groups of ECG traces. (e) The map of the relative activation time of the sinus MI rat. (f) The average activation delay depending on distance for longitudinal (n = 3) and transverse (n = 3) directions of the MI rat. (g) The map of the relative activation time of the ventricular hypertrophy. (h) The average activation delay depending on distance for longitudinal (n = 3) and transverse (n = 3) directions of the ventricular hypertrophy rat. (i) Schematic illustration of the MI cardiac strain beat movement using the MHMEA in a living rat model. (j) Spatiotemporal color mapping of epicardial strain distribution measured using this device platform during a single cardiac cycle of an MI anesthetized rat. (k) Statistical analysis of MI cardiac strain value for the strain sensor array. (l) Schematic illustration of the LVH cardiac strain beat movement using the multimodal device in a living rat model. (m) Spatiotemporal mapping of epicardial strain distribution measured using MHMEA during a single cardiac cycle of an LVH anesthetized rat. (n) Statistical analysis of LVH cardiac strain value for the strain sensor. Data in f, h, k, n are means $\pm$ SD, n = 3. Statistical significance and p values are determined by the ANOVA-test method, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

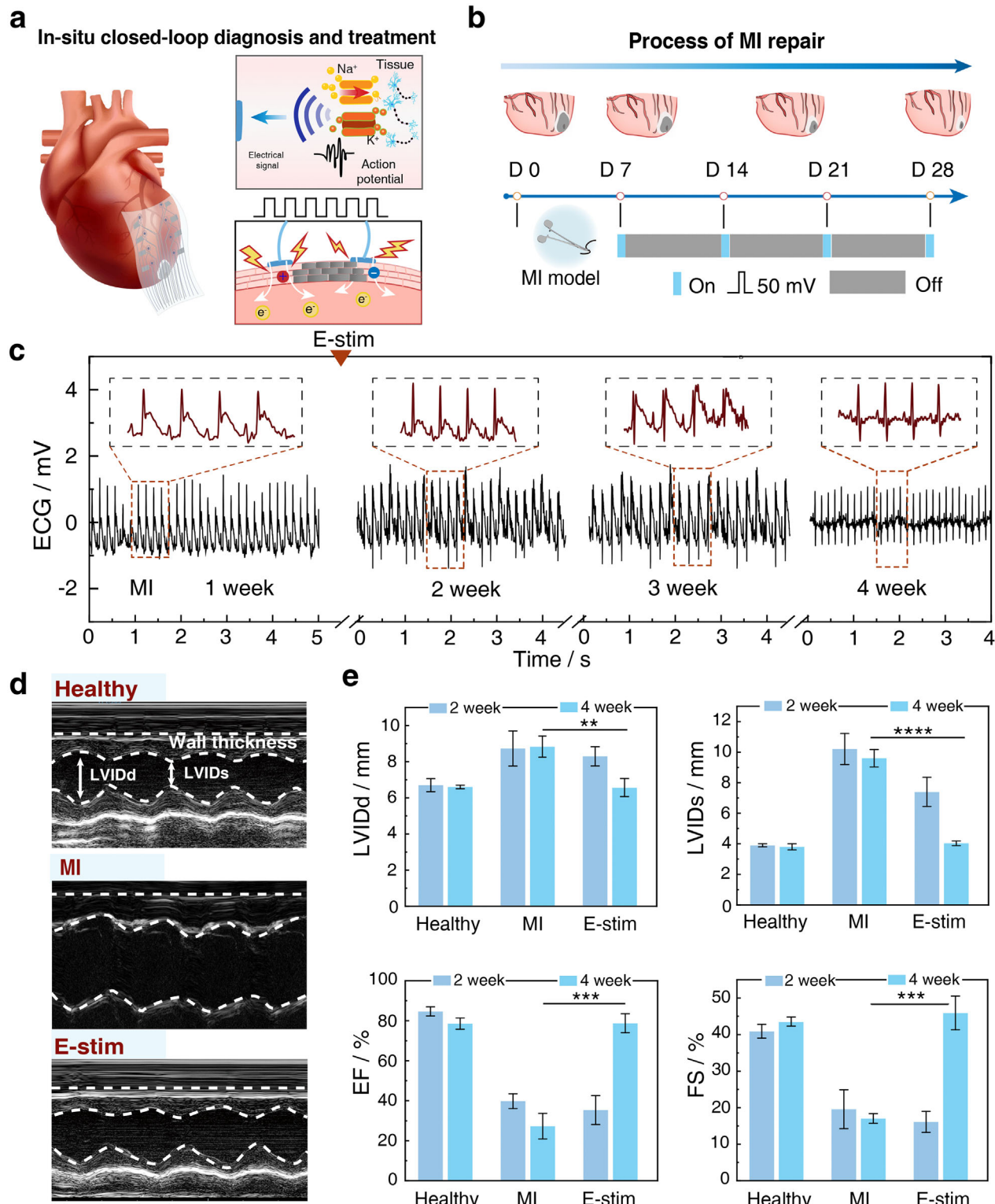


FIGURE 6 | In vivo validation of the MHMEA for MI in situ diagnostic and therapeutic capacity. (a) Schematic illustration in situ monitoring and therapy using the MHMEA in a live rat MI model. (b) Schematic illustration of MHMEA for electrical stimulation-induced MI repair process in rats. (c) Representative measurements of ECG signals for local cardiac contractions using the MHMEA in the process of living rat MI heart repairing. (d) Echocardiographic assessment of left ventricular function in different groups after 4 weeks of electrical stimulation. (e) Statistical analysis of LVIDd, LVIDs, FS, and EF variations among different groups was conducted using echocardiography at 2 and 4 weeks. Data in e are means $\pm$ SD, $n = 3$. Statistical significance and p values are determined by the ANOVA-test method, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. S.D. $n = 3$.

this period is substantially longer than the 4-week therapeutic window. This stable interfacial adhesion to cardiac tissue is thus fundamental to precise electrophysiological modulation. As shown in Figure 6c,d, upon the termination of electrical pacing, the MHMEA captured electrocardiographic signals that reflecting the MI state and its healing process. The heart rate and rhythm returned to normal, transitioning from the typical ST-segment elevation characteristic of MI to normal electrocardiographic signals, indicating the restoration of electrical conduction within the myocardial tissue. Cardiac function was assessed at Week 2 and Week 4 using echocardiography. The echocardiographic images showed pronounced ventricular dilation and reduced motion of the anterior wall of the left ventricle (LV) in the MI group. Quantitative analysis of the echocardiographic data revealed increased left ventricular internal diameter during diastole (LVIDd) and systole (LVIDs), as well as reduced fractional shortening (FS) and ejection fraction (EF), indicating a significant decline in cardiac function in the MI group. In contrast, the Estimote group demonstrated notable improvements in wall motion and cardiac function (Figure 6e). These findings highlight the potential of our hydrogel-based bioelectronic system for precision cardiac therapy, offering a novel approach to the management of cardiac pathologies at the site of abnormal regions.

3 | Conclusion

This study constructs an MHMEA using a multi-material 3D-printing technique, endowing it with tissue-like modulus and superior self-sensing performance, as well as the capacity to interface seamlessly with the dynamic biological tissues and organs. Through the integration of a diverse array of recording, sensing, and stimulating functions into one implantable bioelectronics, it offers the capability to simultaneous spatiotemporal map of the myocardial strain and epicardial electrophysiology in both healthy and diseased rat models and deliver efficient electrical stimulation therapy for diseased areas. This multimodal integration approach not only advances bioelectronic engineering but also establishes novel pathways for personalized cardiac disease management through tissue-device synergy.

4 | Methods

4.1 | Fabrication of the MHMEA

4.1.1 | Preparation of the 3D Printing Ink

Five distinct polymer inks were formulated, each tailored for a specific layer: the adherable hydrogel substrate layer, recording electrode, stimulation electrode, hydrogel strain sensor, and encapsulation layer. The adherable hydrogel substrate was composed of HEC (3.33 wt.%), CTS (0.1 wt.%), PAA-NHS (66.66 wt.%), and GA (0.04 wt.%). The recording electrode layer was composed of PEDOT:PSS dispersible pellets (5 wt.%), HEC (2.6 wt.%), PAA-NHS (21 wt.%), CTS (0.06 wt.%), and GA (0.4 wt.%). The stimulation electrode layer was composed of PEDOT:PSS dispersible pellets (5 wt.%), ethanol (6 wt.%), HEC (2.6 wt.%), PAA-NHS (21 wt.%), and GA (0.4 wt.%). The hydrogel strain sensor layer was composed of PEDOT:PSS dispersible pellets (5 wt.%), HEC (10 wt.%), CTS (1 wt.%), and GA (0.4 wt.%). The

hydrogel encapsulation layer was composed of HEC (10 wt.%) and GA (0.4 wt.%). All inks were mixed evenly using magnetic stirring, followed by degassing through centrifugation (AR-100, Thinky) to form printable inks.

4.1.2 | 3D Printing Procedure

The MHMEA was fabricated utilizing a 3D Bio-Architect multi-nozzle workstation (Regenovo). The intricate print paths were meticulously designed with the aid of commercial software, Adobe Illustrator and SolidWorks (Dassault Systèmes), before being translated into G-code to govern the precise x-y-z movements of the printer heads. Under controlled air pressure, the specialized inks were precision-extruded from 5 mL syringe barrels via nozzles with a diameter of 160 μm . Following the 3D printing procedure, the resultant MHMEA were subjected to a 24-h drying period at ambient temperature to reach a stable dry state. Subsequently, the dried electronics were carefully delaminated from the PET substrate, ensuring the integrity of the structure was maintained. Prior to their application, these electronics were preserved at temperatures below -20°C to ensure long-term stability and functionality.

4.2 | Characterization

4.2.1 | Mechanical Performance

The mechanical tensile properties of the MHMEA, along with their constituent components such as the hydrogel-based recording/stimulating electrodes, hydrogel strain sensors, adhesive layers, and encapsulated hydrogel, were evaluated using a Xin Sansi Universal Testing Machine. Mechanical properties of these hydrogel samples were conducted at a steady deformation rate of 40 mm/min. All hydrogel specimens were thoroughly soaked in PBS prior to the start of testing, ensuring they were fully hydrated.

4.2.2 | Electrical Performance

For the electrical conductivity measurements, the hydrogel samples were cut into regular strips using a laser cutter (Corelaser, K1032). The electrical resistance of hydrogel samples was tested using an LCR (Tonghui, TH2829C) instrument. The conductivity of the samples was calculated as:

$$\sigma = L / (R \times W \times T)$$

where L is the distance between the two electrodes for resistance measurement, W and T are the width and thickness of the sample, respectively, and R is the resistance of the sample.

The strain-sensing performance of the hydrogel strain sensors was quantitatively assessed using a combination of an LCR meter and a mechanical stretching apparatus (Cellscale). This setup enabled the precise measurement of electrical resistance variations of hydrogel strain sensors under varying mechanical strains, as well as the dynamic tracking of relative electrical resistance changes during cyclic deformation tests.

Cyclic voltammetry (CV) and electrical impedance spectroscopy (EIS) measurements of the hydrogel electrodes were performed with a three-electrode system by using the Electrochemical Workstation (CHI660E electrochemical analyzer). All the electrochemical measurements were measured in PBS buffer, while a carbon cloth electrode modified with conducting polymer hydrogel as the working electrode, a bare Pt sheet as a counter electrode, and an Ag/AgCl electrode as the reference electrode, respectively. For charge injection performance (CIC), electrochemical current pulse injection (in recurrent potential pulses mode, -0.5 – 0.5 V vs. Ag/AgCl) tests were carried out by using an electrochemical workstation (Gamry Instruments).

4.3 | Animal Experiments

All animal procedures used were strictly compliant with the standard guidelines approved by the Southern University of Science and Technology Ethics Committee (SUSTech-JY202308104). 6-week-old, Sprague Dawley (SD) rats weighing 200–300 g were used in this study. Animals were housed in controlled conditions (ambient temperature $24^{\circ}\text{C} \pm 1^{\circ}\text{C}$, humidity 50%–60%, 12-h light/dark cycle) with free access to food and water.

SD rats were anesthetized with tribromoethanol, mechanically immobilized, and intubated for tracheal intubation. Mechanical ventilation was provided using a ventilator (RWD, R415). After full anesthesia, a thoracotomy was performed through the third or fourth intercostal space on the left side to expose the heart. The pericardium was carefully removed with small forceps, and most biological fluids were absorbed with gauze. After the left lateral thoracotomy, the left anterior descending coronary artery (LAD) was ligated 2–3 mm below the pulmonary artery cone and left atrium using a 7-0 suture. Successful ligation was confirmed by ST-segment elevation on the electrocardiogram and the blanching of the ligated area due to ischemia. For the left ventricular hypertrophy rat model, rats received intraperitoneal injections of thyroid hormone (1 mg/kg) every 24 h for 7 consecutive days. The high dose of thyroxine increased basal metabolic rate, enhanced sympathetic activity, and increased cardiac workload. Elevated levels of thyroxine in vivo promoted mRNA and protein synthesis in cardiomyocytes, leading to myocardial hypertrophy. SD rats (200–300 g) were anesthetized with tribromoethanol and intubated. Following thoracotomy, an 8-channel hydrogel electrode array was gently applied to the anterior surface of the heart, and a 0.5 mm diameter metal wire was inserted subcutaneously in the leg as a reference electrode. After the procedure, the animals were humanely euthanized using concentrated pure carbon dioxide.

4.3.1 | In vivo Electrophysiological Recording

SD rats were anesthetized with tribromoethanol and mechanically fixed, intubated and connected to a mechanical ventilator (RWD, R415). After general anesthesia, the rat chest cavity was opened by thoracotomy in the left third or fourth intercostal region to expose the heart. The pericardium was removed from the rat heart with small forceps, and most of the biological fluid was removed with gauze. The MHMEA was then carefully placed onto the ventricular surface, covering the targeted

region. A 0.5-mm diameter wire was inserted subcutaneously into the rat's leg to serve as a reference electrode. The connecting wire of the MHMEA was subsequently connected to a multi-channel electrophysiological instrument (XDAQ Multi-Channel Electrophysiology Stimulation/Acquisition System). This integrated configuration not only enabled continuous recording of epicardial electrograms and real-time monitoring of cardiac electrical activity, but also allowed for the precise delivery of targeted electrical stimulation to modulate abnormal cardiac rhythms.

4.3.2 | Rat Model of MI and E-Stim Treatment

The myocardial infarction (MI) model was established by permanent ligation of the left anterior descending coronary artery (LAD). Briefly, the LAD was ligated 2–3 mm distal to the pulmonary artery cone and left atrium using a 7-0 polypropylene suture. Successful coronary occlusion was verified by immediate ST-segment elevation on electrocardiography and visible pallor of the anterior ventricular wall indicative of regional myocardial ischemia. Subsequently, the MHMEA was placed onto the epicardial surface covering the infarcted region. 7 days after ligation, when the MI was fully established and the infarct region had stabilized, intermittent electrical stimulation was initiated at predetermined time points (once every 7 days post-MI, i.e., on days 7, 14, and 21). For electrical stimulation, targeted electrical stimulation (50 mV, 8 Hz, 4 ms pulse duration) was applied for 4 h per session in an intermittent mode to modulate abnormal cardiac activity, following previously reported protocols [50–52].

4.3.3 | Histocompatibility Assessment

Histological analysis was carried out to evaluate the inflammation response. This MHMEA were implanted into the subcutaneous tissue of the backs of mice for 30 days. Biological tissue samples collected from the animal models were fixed with 4 vol.% paraformaldehyde solution, followed by sequential dehydration using ethanol gradients (75%–90%–95%–100%), and subsequently embedded in paraffin. The sample sectioning was conducted with a Leica RM2016 Cryostat (Leica, Germany) into slides of 4 μm in thickness, followed by hematoxylin and eosin (H&E) and Masson trichrome staining. Histological analysis was performed by H&E (DC0032, Leagene Biotechnology) staining according to standard protocols.

4.4 | Simulation Analysis

4.4.1 | 3D FEM

The structural stability of the bioelectronic device under tensile strain was investigated using a 3D finite element method (FEM). The geometric model was imported into the preprocessor of the ABAQUS finite element software. The four-node linear tetrahedral solid element (C3D4) and four-node quadrilateral membrane element (M3D4) in ABAQUS were used for modeling. The total number of elements exceeded 60 000, and mesh refinement ensured the accuracy of the numerical results.

4.4.2 | Statistical Analysis

All results are presented as the mean $\pm$ standard deviation (S.D.). All experiments were repeated at least three times, and each condition was analyzed in triplicate. One-way analysis of variance (one-way ANOVA) followed by Tukey's post-hoc test was used to evaluate statistical significance among multiple groups. A significance level of $p < 0.05$ was considered statistically significant. All statistical analyses were performed using Origin and MATLAB software package.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm75482-sup-0001-SuppMat.pdf.